

课后习题(5) - 语法制导翻译

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1. (1) 用 int 和 real 分别表示整型和实型数.

属性文法如下:

$$E \rightarrow E_1 * T \quad \{ \text{if } E_1.\text{type} = \text{int} \text{ and } T.\text{type} = \text{int} \\ \text{then } E.\text{type} := \text{int} \\ \text{else } E.\text{type} := \text{real} \}$$

$$E \rightarrow T \quad \{ E.\text{type} := T.\text{type} \}$$

$$T \rightarrow \text{num}.\text{num} \quad \{ T.\text{type} := \text{real} \}$$

$$T \rightarrow \text{num} \quad \{ T.\text{type} := \text{int} \}$$

(2) 其中 lexeme 表示单词拼写. int-to-real 表示整型数转换为实型数.

$$E \rightarrow E_1 * T \quad \{ \text{if } E_1.\text{type} = \text{real} \text{ and } T.\text{type} = \text{int} \text{ then} \\ \text{begin} \\ E.\text{type} := \text{real}; \text{print}(T.\text{lexeme}); \text{print}(' \text{int-to-real} '); \\ \text{end.} \\ \text{else if } E_1.\text{type} = \text{int} \text{ and } T.\text{type} = \text{real} \text{ then} \\ \text{begin} \\ E.\text{type} := \text{real}; \text{print}(T.\text{lexeme}); \text{print}(' \text{int-to-real} '); \\ \text{end} \\ \text{else begin.} \\ E.\text{type} := E_1.\text{type}; \text{print}(T.\text{lexeme}); \\ \text{end} \\ \text{print}('*'); \}$$

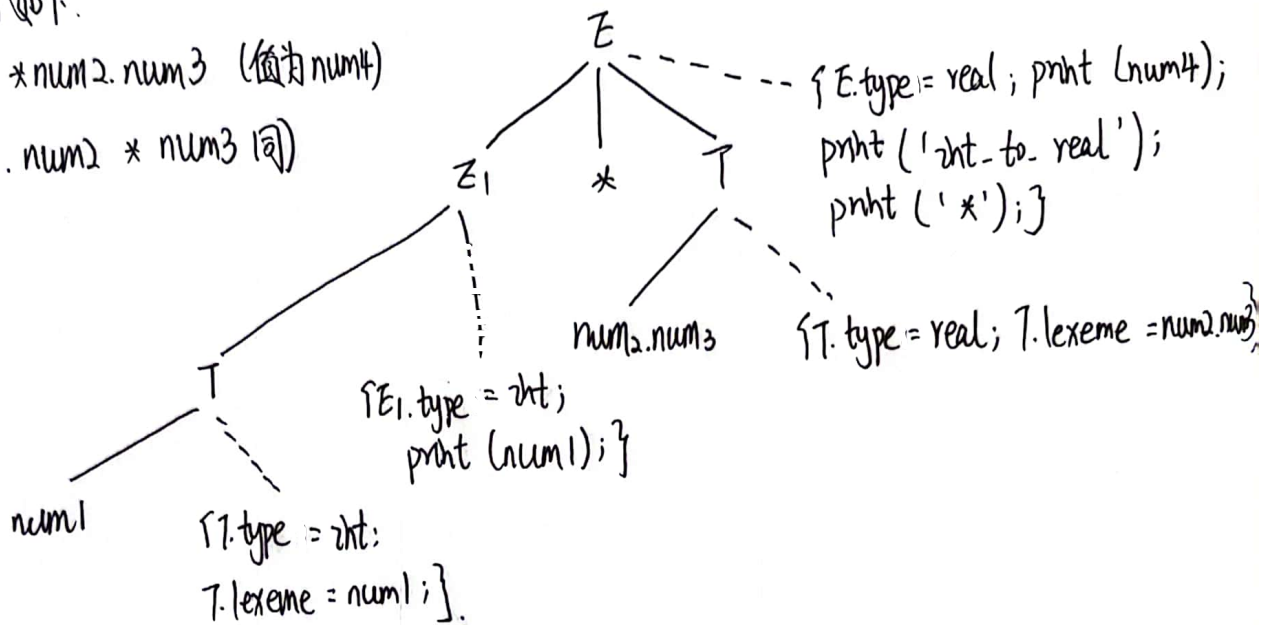
$$E \rightarrow T \quad \{ E.\text{type} := T.\text{type}; \text{print}(T.\text{lexeme}); \}$$

$$T \rightarrow \text{num}.\text{num} \quad \{ T.\text{type} := \text{real}; T.\text{lexeme} := \text{num1}.\text{lexeme} \parallel "." \parallel \text{num2}.\text{lexeme} \}$$

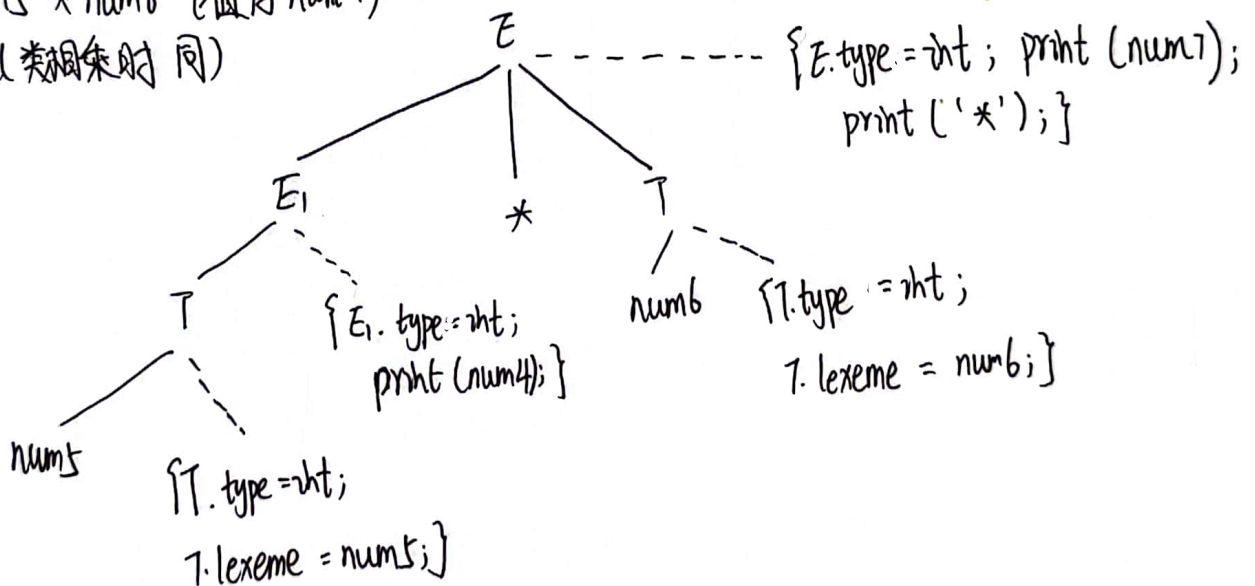
$$T \rightarrow \text{num} \quad \{ T.\text{type} := \text{int}; T.\text{lexeme} := \text{num}.\text{lexeme}; \}$$

语义树如下。

- ① $num1 * num2.num3$ (值为 $num4$)
(与 $num1.num2 * num3$ 同)



- ② $num5 * num6$ (值为 $num7$)
(两个 $real$ 类相乘时同)



2. 解: 综合属性 $length$ 表示二进制的位数

产生式
 $S \rightarrow L_1.L_2.$

语义规则
 $\{ S.val := L_1.val + L_2.val * 2^{-L_2.length}; \}$

$S \rightarrow L$

$\{ S.val := L.val \}$

$L \rightarrow L_1B$

$\{ L.val := L_1.val * 2 + B.bval; \\ L.length := L_1.length + 1 \}$

$L \rightarrow B$

$\{ L.val := B.bval; L.length := 1; \}$

$B \rightarrow 0$

$\{ B.bval := 0 \}$

$B \rightarrow 1$

$\{ B.bval := 1 \}$

3. 解: 翻译模式如下:

$$L \rightarrow id L \{ addtype (id.entry, L.type) \}$$
$$L \rightarrow , id L \{ addtype (id.entry, L.type) \}$$
$$L \rightarrow : T \{ L.type = T.type \}$$
$$T \rightarrow integer \{ T.type = integer \}$$
$$T \rightarrow real \{ T.type = real \}$$